

Education

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2024 – Present

PhD, Computing

University of Michigan–Flint · Audio Deepfake Detection, Signal Processing

2022 – 2024

PhD, Computer Science

Oakland University, Rochester, MI · Audio Deepfake Detection

2017 – 2020

MS, Computer Science

University of Engineering & Technology, Taxila, Pakistan · Thesis: Blur Image Region Detection and Segmentation

2013 – 2017

BS, Computer Science

University of Wah, Pakistan · Thesis: Performance Analysis of 802.11 Protocol

## Experience

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2026 – 2027

### AI Developer Intern

*ProbeTruth, Inc. · Flint, Michigan*

- Designed, developed, and tested AI/ML models for real-world applications using Python and modern ML frameworks
- Built and optimized data pipelines for preprocessing, feature extraction, and model training
- Worked with multimodal data (text, audio, image, video) for AI-driven detection solutions

2024 – 2026

### Research Assistant

*University of Michigan–Flint · SMILES Lab*

- Interpretable deepfake detection using symbolic and feature-based learning
- ML pipelines for large-scale audio datasets
- NSF-funded projects on multimodal deepfake analysis

2017 – 2020

### Research Assistant (fully funded)

*University of Engineering & Technology, Taxila, Pakistan*

- Machine learning and computer vision research: feature design and statistical modeling
- Developed the Local Directional Mean Pattern (LDMP) feature extraction method

## Publications

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19. Dual-Branch Architecture for Audio Deepfake Detection.

[A. Khan](#), K. M. Malik. *INTERSPEECH — Main Track, 2026*. first author accepted

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18. TRACE: Training-Free Partial Audio Deepfake Detection via Embedding Trajectory Analysis of Speech Foundation Models.

[A. Khan](#), K. M. Malik. *CVPR Workshop — MMFM, 2026*. first author accepted

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17. PSA-Net: Parallel Stacked Aggregated Network for Voice Authentication in IoT-Enabled Smart Devices.

[A. Khan](#), I. U. Haq, K. M. Malik. *IEEE Trans. Consumer Electronics, 2026*. first author accepted

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16. Interpretable and Generalizable Audio Deepfake Detection using Joint Spectro-Temporal and Neuro-symbolic Learning.

[A. Khan](#), K. M. Malik. *IEEE Trans. Information Forensics & Security, 2026*. first author under review

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15. Adversarial Attacks on Audio Deepfake Detection: A Benchmark and Comparative Study.

K. Uddin, M. U. Farooq, [A. Khan](#), K. M. Malik. *arXiv:2509.07132, 2025*.

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14. SHIELD: A Secure and Highly Enhanced Integrated Learning for Robust Deepfake Detection Against Adversarial Attacks.

K. Uddin, [A. Khan](#), M. U. Farooq, K. M. Malik. *arXiv:2507.13170, 2025*.

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13. Generalized Deepfake Detection Using Identity, Behavioral, and Geometric Signatures.

M. U. Farooq, [A. Khan](#), I. U. Haq, K. M. Malik. *IEEE Trans. Computational Social Systems, 2025*.

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12. Transferable Adversarial Attacks on Audio Deepfake Detection.

M. U. Farooq, [A. Khan](#), K. Uddin, K. M. Malik. *arXiv:2501.11902, 2025*.

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11. Securing Social Media Against Deepfakes Using Identity, Behavioral, and Geometric Signatures.

M. U. Farooq, [A. Khan](#), I. U. Haq, K. M. Malik. *arXiv:2412.05487, 2024*.

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10. Bridging the Spoof Gap: A Unified Parallel Aggregation Network for Voice Presentation Attacks.

[A. Khan](#), K. M. Malik. *Engineering Applications of AI · IF 8.5, 2023*. first author in process

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9. Securing Voice Biometrics: One-Shot Learning Approach for Audio Deepfake Detection.

[A. Khan](#), K. M. Malik. *IEEE WIFS, 2023*. first author

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8. SpotNet: A Spoofing-Aware Transformer Network for Effective Synthetic Speech Detection.

**A. Khan**, K. M. Malik. *ACM-MAD*, 2023. first author

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7. **Frame-to-Utterance Convergence: A Spectra-Temporal Approach for Unified Spoofing Detection.**

**A. Khan**, K. M. Malik, S. Nawaz. *IEEE ICASSP*, 2023. first author

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6. **Battling Voice Spoofing: A Review, Comparative Analysis, and Generalizability Evaluation of State-of-the-Art Voice Spoofing Countermeasures.**

**A. Khan**, K. M. Malik, J. Ryan, M. Saravanan. *Artificial Intelligence Review*, 56(Suppl. 1) · IF 12.0, 2023.

first author

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5. **Toward Realigning Automatic Speaker Verification in the Era of COVID-19.**

**A. Khan**, A. Javed, K. M. Malik, M. A. Raza, J. Ryan, A. K. J. Saudagar, H. Malik. *Sensors*, 22(7), 2638 · IF 3.85, 2022. first author

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4. **A Robust Approach for Blur and Sharp Regions' Detection Using Multisequential Deviated Patterns.**

**A. Khan**, A. Javed, A. Irtaza, M. T. Mahmood. *International Journal of Optics* · Q2, 2021. first author

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3. **Defocus Blur Detection Using Novel Local Directional Mean Patterns (LDMP) and Segmentation via KNN Matting.**

**A. Khan**, A. Irtaza, A. Javed, T. Nazir, H. Malik, K. M. Malik, M. A. Khan. *Frontiers of Computer Science*, 16(2) · Q1, 2020. first author

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2. **Segmentation of Defocus Blur Using Local Triplicate Co-Occurrence Patterns (LTCOP).**

**A. Khan**, S. A. Irtaza, A. Javed, M. A. Khan. *MACS-13*, 2019. first author

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1. **Detection of Blur and Non-Blur Regions Using Frequency-Based Multi-Level Fusion Transformation and Classification via KNN Matting.**

M. A. Khan, S. A. Irtaza, **A. Khan**. *MACS-13*, 2019.

Full list with links on the [Publications](#) page.

## Skills

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### PROGRAMMING

Python, MATLAB, PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

### AUDIO & SIGNAL

Librosa, Praat, spectral analysis, handcrafted feature extraction

### ML / DL

CNN, RNN, transformer architectures — model design, training, evaluation

### TOOLS

Git, LaTeX, Jupyter, Google Colab, virtual machines

## Awards & Scholarships

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- Fully Funded Research Scholar — University of Michigan–Flint
- HEC Pakistan PhD Scholarship — Higher Education Commission, Pakistan
- Fully Funded Master's Scholarship — UET Taxila
- Appreciation Award — MACS-13 Conference
- 1st Position — PBTE Lahore board examinations

## Referees

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### Dr. Khalid Mahmood Malik

*Professor of Computer Science, University of Michigan · drmalik@umich.edu*

### Dr. Syed Aun Irtaza

*Associate Professor, UET Taxila · aun.irtaza@uettaxila.edu.pk*